

# Practical 1 – The Fixed Voltage Divider

Booklet 3, section 9, page 21. Success criteria 4.1, 4.2 and 2.6.

Everything you write goes in your jotter. Copy each numbered table before you start that section, and answer each question by its number.

## The idea being tested

Two resistors in series share the supply between them, in the same ratio as their resistances. You are going to find out how closely the real circuit obeys that rule.

$V_1 / V_2 = R_1 / R_2$  – it is on the data sheet, so you never have to recall it

## 1.1 What you are testing

**Aim:** to find out whether two resistors in series share the supply voltage in the same ratio as their resistances.

You will test four pairs of resistors on a 5 V supply. Three of them make a point on their own; the fourth is an extreme case.

| Pair | $R_1$ and $R_2$                 | Why this pair is in the list                       |
|------|---------------------------------|----------------------------------------------------|
| A    | 1 k $\Omega$ and 1 k $\Omega$   | equal resistors – the simplest case                |
| B    | 1 k $\Omega$ and 2.2 k $\Omega$ | uneven – which one takes the bigger share?         |
| C    | 2.2 k $\Omega$ and 1 k $\Omega$ | pair B swapped over – does the answer swap too?    |
| D    | 1 k $\Omega$ and 10 k $\Omega$  | one resistor ten times the other – an extreme case |

## 1.2 Predict

Table 1.2 – predicted ratio

| Pair | $R_1$          | $R_2$          | $R_1 / R_2$ |
|------|----------------|----------------|-------------|
| A    | 1 k $\Omega$   | 1 k $\Omega$   |             |
| B    | 1 k $\Omega$   | 2.2 k $\Omega$ |             |
| C    | 2.2 k $\Omega$ | 1 k $\Omega$   |             |
| D    | 1 k $\Omega$   | 10 k $\Omega$  |             |

**Q1** Before you build anything: which resistor do you expect to have the larger voltage across it – the larger one or the smaller one? Say why. [2]

1.3 Simulate in Yenka

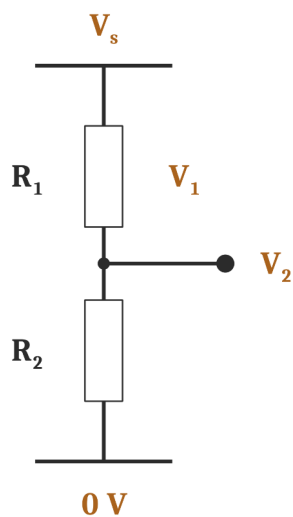


Table 1.3 – simulated readings

| Pair | $V_1$ (V) | $V_2$ (V) | $V_1 + V_2$ (V) | $V_1 / V_2$ |
|------|-----------|-----------|-----------------|-------------|
| A    |           |           |                 |             |
| B    |           |           |                 |             |
| C    |           |           |                 |             |
| D    |           |           |                 |             |

- Q2 Look down your  $V_1 + V_2$  column. Write what you notice as a rule linking  $V_1$ ,  $V_2$  and the supply voltage  $V_s$ , and explain why it happens.

Q3 Use your rule to find  $V_1$  when  $V_2$  is 3.2 V on a 5 V supply.
- [2]

[1]

1.4 Build on the breadboard

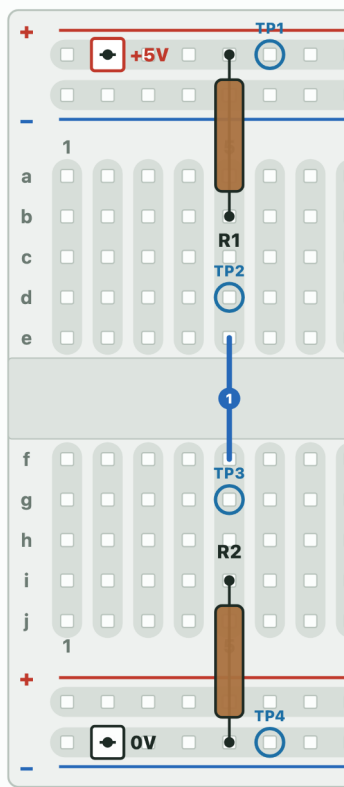


Table 1.4 – measured readings

| Pair | $V_1$ (V) | $V_2$ (V) | $V_1 / V_2$ | Meter range |
|------|-----------|-----------|-------------|-------------|
| A    |           |           |             |             |
| B    |           |           |             |             |
| C    |           |           |             |             |
| D    |           |           |             |             |

Measure  $V_1$  between TP1 and TP2, and  $V_2$  between TP3 and TP4.

Pair A.  $R_1$ , the jumper and  $R_2$  run down column 5 from the +5 V rail to the 0 V rail.

## 1.5 Compare

Copy your ratios from Tables 1.2, 1.3 and 1.4. The last column is the difference between what you predicted and what you measured.

Table 1.5 – the three answers side by side

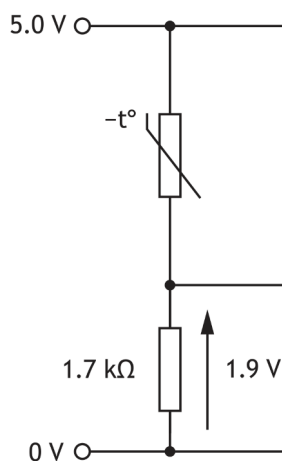
| Pair | Predicted $R_1 / R_2$ | Simulated $V_1 / V_2$ | Measured $V_1 / V_2$ | Measured – predicted |
|------|-----------------------|-----------------------|----------------------|----------------------|
| A    |                       |                       |                      |                      |
| B    |                       |                       |                      |                      |
| C    |                       |                       |                      |                      |
| D    |                       |                       |                      |                      |

- Q4** Which two of the three columns agree most closely? Suggest why. [2]
- Q5** Give two reasons why the measured ratio is not exactly the predicted one. Neither reason is 'I made a mistake'. [2]
- Q6** Use your measured  $V_1$  and  $V_2$  for pair B, and  $R_1 = 1 \text{ k}\Omega$ , to calculate  $R_2$ . Compare your answer with the printed  $2.2 \text{ k}\Omega$ . [3]
- Q7** A resistor's printed value has a tolerance – often  $\pm 5\%$ . Explain how that alone could account for the difference you found in Q6. [2]

## 1.6 What the results tell you

- Q8** Look at pairs B and C. State what happened to  $V_2$  when the two resistors were swapped over, and explain why. [3]
- Q9** In pair D, almost all of the supply appeared across one resistor. Name it and explain why it takes the larger share. [3]
- Q10** A divider is built from a  $1\text{ k}\Omega$  and a  $1\text{ M}\Omega$  resistor across a  $9\text{ V}$  supply, with the  $1\text{ M}\Omega$  as the lower resistor. State which resistor has almost all of the  $9\text{ V}$  across it, and explain why using the ratio of their resistances. [2]
- Q11** State the name of this type of series circuit. [1]

From a past exam paper



This input sub-system is from a laminator. The top component is a thermistor: a resistor whose resistance changes with temperature.

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Scotland (SQA). National 5 Engineering  
Science 2023, Question 14(b).

- Q12** Calculate the resistance of the thermistor for the condition shown. [4]